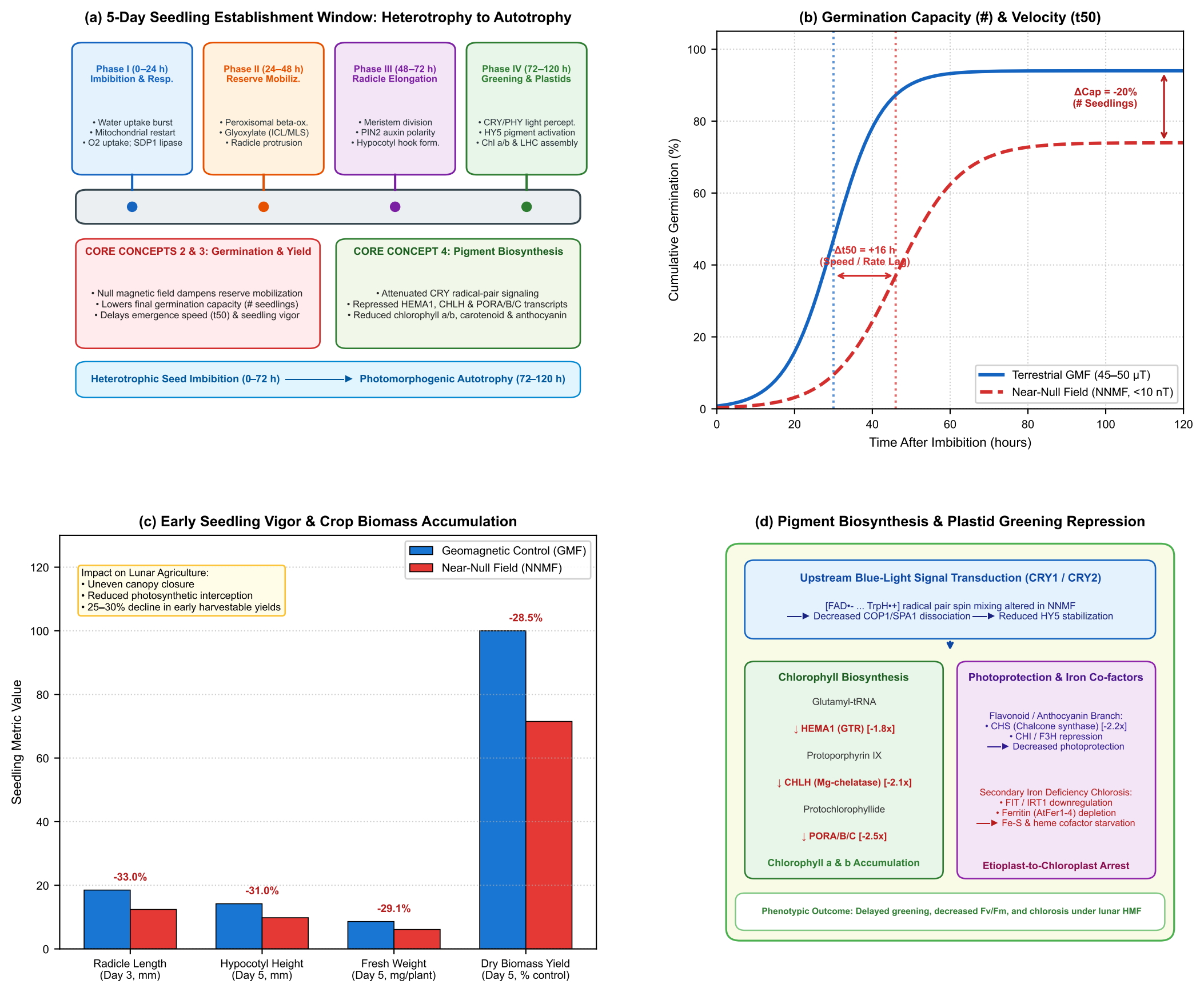

(a) 5-Day Seedling Establishment Window: Heterotrophy to Autotrophy
Phase I (0–24 h)
Imbibition & Resp.
• Water uptake burst
• Mitochondrial restart
• O2 uptake; SDP1 lipase
Phase II (24–48 h)
Reserve Mobiliz.
• Peroxisomal beta-ox.
• Glyoxylate (ICL/MLS)
• Radicle protrusion
Phase III (48–72 h)
Radicle Elongation
• Meristem division
• PIN2 auxin polarity
• Hypocotyl hook form.
Phase IV (72–120 h)
Greening & Plastids
• CRY/PHY light percept.
• HY5 pigment activation
• Chl a/b & LHC assembly
CORE CONCEPTS 2 & 3: Germination & Yield
• Null magnetic field dampens reserve mobilization
• Lowers final germination capacity (# seedlings)
• Delays emergence speed (t50) & seedling vigor
CORE CONCEPT 4: Pigment Biosynthesis
• Attenuated CRY radical-pair signaling
• Repressed HEMA1, CHLH & PORA/B/C transcripts
• Reduced chlorophyll a/b, carotenoid & anthocyanin
Heterotrophic Seed Imbibition (0–72 h)
Photomorphogenic Autotrophy (72–120 h)
(b) Germination Capacity (#) & Velocity (t50)
Cumulative Germination (%)
Time After Imbibition (hours)
ΔCap = -20%
(# Seedlings)
Δt50 = +16 h
(Speed / Rate Lag)
Terrestrial GMF (45–50 μT)
Near-Null Field (NNMF, <10 nT)
(c) Early Seedling Vigor & Crop Biomass Accumulation
Seedling Metric Value
Geomagnetic Control (GMF)
Near-Null Field (NNMF)
Impact on Lunar Agriculture:
• Uneven canopy closure
• Reduced photosynthetic interception
• 25–30% decline in early harvestable yields
-33.0%
-31.0%
-29.1%
-28.5%
Radicle Length
(Day 3, mm)
Hypocotyl Height
(Day 5, mm)
Fresh Weight
(Day 5, mg/plant)
Dry Biomass Yield
(Day 5, % control)
(d) Pigment Biosynthesis & Plastid Greening Repression
Upstream Blue-Light Signal Transduction (CRY1 / CRY2)
[FAD•- ... TrpH•+] radical pair spin mixing altered in NNMF
Decreased COP1/SPA1 dissociation
Reduced HY5 stabilization
Chlorophyll Biosynthesis
Glutamyl-tRNA
↓ HEMA1 (GTR) [-1.8x]
Protoporphyrin IX
↓ CHLH (Mg-chelatase) [-2.1x]
Protochlorophyllide
↓ PORA/B/C [-2.5x]
Chlorophyll a & b Accumulation
Photoprotection & Iron Co-factors
Flavonoid / Anthocyanin Branch:
• CHS (Chalcone synthase) [-2.2x]
• CHI / F3H repression
Decreased photoprotection
Secondary Iron Deficiency Chlorosis:
• FIT / IRT1 downregulation
• Ferritin (AtFer1-4) depletion
Fe-S & heme cofactor starvation
Etioplast-to-Chloroplast Arrest
Phenotypic Outcome: Delayed greening, decreased Fv/Fm, and chlorosis under lunar HMF